

Hostname (Master): hpc-master Domain: hpc.local

1. Virtual Cluster Architecture

Node Role	Hostname	IP Address	MAC Address	OS Install Method
Master / Head Node	hpc-master	192.168.179.140	—	Manual
Login Node	login	192.168.100.11	00:50:56:34:93:8D	xCAT
Compute Node 1	compute1	192.168.100.13	00:50:56:3A:18:A7	xCAT
Compute Node 2	compute2	192.168.100.14	00:50:56:2C:29:67	xCAT
Storage Node	storage	192.168.100.12	00:50:56:26:46:3D	Manual

2. xCAT Installation Verification

2.1 xCAT Version

Command:

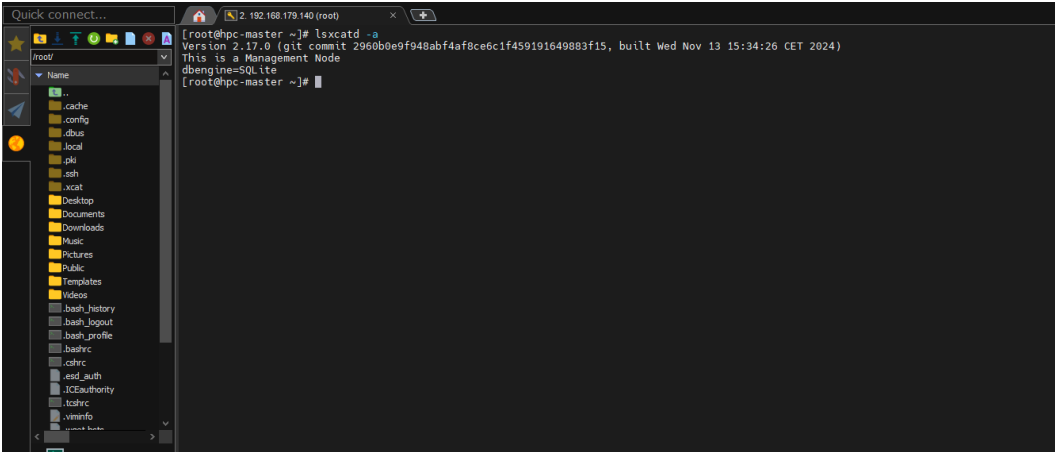
```
lsxcatd -a
```

Output:

```
Version 2.17.0 (git commit 2960b0e9f948abf4af8ce6c1f459191649883f15, built Wed Nov 13 15:34:26 CET 2024)
This is a Management Node
dbengine=SQLite
```

Observations:

- xCAT version 2.17.0 is installed on the Master node.
- The node is correctly identified as a Management Node.
- The database engine in use is SQLite (suitable for small to medium clusters).



2.2 xCAT Service Status

Command used:

```
systemctl status xcatd --no-pager
```

Output summary:

- xcatd.service - xCAT management service
Loaded: loaded (/usr/lib/systemd/system/xcatd.service; enabled; vendor preset: disabled)
Active: active (running) since Mon 2026-03-02 10:05:26 IST; 28min ago
Main PID: 5562
Tasks: 7 (limit: 10810)
Memory: 91.8M

Observations:

- xcatd is running and enabled (will auto-start on reboot).
- All sub-processes are healthy.
- The install monitor is actively tracking node provisioning events.

```
dpeng@hpc-master:~$ systemctl status xcatd --no-pager
● xcatd.service - xCAT management service
   Loaded: loaded (/usr/lib/systemd/system/xcatd.service; enabled; vendor preset: disabled)
   Active: active (running) since Mon 2026-03-02 10:05:26 IST; 28min ago
     Process: 1227 ExecStart=/etc/init.d/xcatd start (code=exited, status=0/SUCCESS)
    Main PID: 5562 (xcatd: SSL list)
      Tasks: 7 (limit: 10810)
     Memory: 91.8M
    CGroup: /system.slice/xcatd.service
            └─5559 /usr/sbin/in.tftpd -v -l -s /tftpboot -m /etc/tftpmmapfile4xcat.conf
              └─5562 xcatd: SSL listener
                └─5568 xcatd: DB Access
                  └─5576 xcatd: UDP listener
                    └─5580 xcatd: install monitor
                      └─5582 xcatd: Command log writer
                        └─5583 xcatd: Discovery worker

Mar 02 10:27:21 hpc-master.hpc.local xcat[5580]: xcatd: finish a connection request for compute2 from 192.168.100.14
Mar 02 10:29:57 hpc-master.hpc.local xcat[5580]: xcatd: install monitor received a connection request from 192.168.100.11
Mar 02 10:29:57 hpc-master.hpc.local xcat[5580]: xcatd: 192.168.100.11 is matched with node login
Mar 02 10:29:57 hpc-master.hpc.local xcat[5580]: xcat.updatestatus - login: changing status=booted
Mar 02 10:29:57 hpc-master.hpc.local xcat[5580]: xcatd: finish a connection request for login from 192.168.100.11
Mar 02 10:31:34 hpc-master.hpc.local xcat[5580]: xcatd: install monitor received a connection request from 192.168.100.12
Mar 02 10:31:34 hpc-master.hpc.local xcat[5580]: xcatd: 192.168.100.12 is matched with node storage
Mar 02 10:31:35 hpc-master.hpc.local xcat[5580]: xcat.updatestatus - storage: changing status=booted
Mar 02 10:31:35 hpc-master.hpc.local xcat[5580]: xcatd: finish a connection request for storage from 192.168.100.12
Mar 02 10:32:17 hpc-master.hpc.local xcat[15458]: xCAT: Allowing lxxcatd -a for root from localhost
[root@hpc-master ~]#
```

3. Node Definitions & Network Configuration

3.1 Defined Nodes

Command used:

```
lsdef -t node
```

Output:

```
compute1    (node)
compute2    (node)
hpc-master  (node)
login       (node)
storage     (node)
```

All 5 nodes are correctly defined in xCAT's database.

3.2 Node Groups

Command used:

```
lsdef -t group
```

Output:

```
__mgmtnode  (group)
all         (group)
compute     (group)
login       (group)
storage     (group)
```

Observations:

- The `compute` group contains `compute1` and `compute2`, enabling group-wide commands.
 - The `all` group covers all managed nodes.
 - `__mgmtnode` is the internal xCAT group for hpc-master itself.
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3.3 Network Configuration

Command used:

tabdump networks

Three networks defined:

Network	Subnet Mask	Interface	Gateway / xCAT Master
192.168.100.0	255.255.255.0	ens224	192.168.100.1
192.168.122.0	255.255.255.0	virbr0	xcatmaster
192.168.179.0	255.255.255.0	ens160	xcatmaster

- The **192.168.100.0/24** network is the primary cluster provisioning network with DHCP range 192.168.100.200–192.168.100.250 and domain hpc.local.

3.4 MAC Address Table

Command used:

tabdump mac

Output:

Node	MAC Address
login	00:50:56:34:93:8D
compute1	00:50:56:3A:18:A7
compute2	00:50:56:2C:29:67
storage	00:50:56:26:46:3D

Observations:

- MAC addresses are recorded for all 4 managed nodes, which is required for DHCP assignment and PXE boot during provisioning.

```
[root@hpc-master ~]# lsdef -t node
compute1 (node)
compute2 (node)
hpc-master (node)
login (node)
storage (node)
[root@hpc-master ~]#
[root@hpc-master ~]#
[root@hpc-master ~]# lsdef -t group
mgmtnode (group)
all (group)
compute (group)
login (group)
storage (group)
[root@hpc-master ~]#
[root@hpc-master ~]#
[root@hpc-master ~]# lsdef -t site | head -40
clustersite (site)
[root@hpc-master ~]#
[root@hpc-master ~]# tabdump networks
#netname,net,mask,mgtifname,gateway,dhcpserver,tftpserver,ntpservers,logservers,dynamicrange,staticrange,staticrangeincrement,n
odehostname,ddnsdomain,vlanid,domain,mtu,comments,disable
"192.168.100.0-255.255.255.0","192.168.100.0","255.255.255.0","ens224","192.168.100.1","192.168.100.1","192.168.100.1",,,,"1
92.168.100.200-192.168.100.250",,,,"hpc.local","1500",,
"192.168.122.0-255.255.255.0","192.168.122.0","255.255.255.0","virbr0","xcatmaster>","xcatmaster>","1500",,
"192.168.179.0-255.255.255.0","192.168.179.0","255.255.255.0","ens160","192.168.179.2","xcatmaster>","1500",,
[root@hpc-master ~]#
[root@hpc-master ~]# tabdump mac
#node,interface,mac,comments,disable
"login","00:50:56:34:93:8D",,
"compute1","00:50:56:3A:18:A7",,
"compute2","00:50:56:2C:29:67",,
"storage","00:50:56:26:46:3D",,
```

4. Provisioning Verification

4.1 Available OS Images

Command used:

```
lsdef -t osimage
```

Output:

```
rocky8.10-x86_64-install-compute      (osimage)
rocky8.10-x86_64-install-service      (osimage)
rocky8.10-x86_64-netboot-compute      (osimage)
rocky8.10-x86_64-stateful-mgmtnode    (osimage)
rocky8.10-x86_64-statelite-compute    (osimage)
```

Observations:

- Rocky Linux 8.10 images are available for compute, service, netboot, and statelite provisioning modes.
 - The `install-compute` image was used to provision compute1, compute2, and login.
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4.2 Node Provisioning Status

Command used:

```
tabdump nodelist
```

Output:

Node	Group	Status
login	login, all	booted
compute1	compute, all	booted
compute2	compute	booted
storage	storage	booted
hpc-master	__mgmtnode	—

Observations:

- All provisioned nodes (compute1, compute2, login) show status **booted**, confirming provisioning completed successfully.
- Storage node also shows booted, consistent with it being manually installed and later registered in xCAT.
- hpc-master has no install status as it is the management node itself.

```
2. 192.168.179.140 (root)
[root@hpc-master ~]# #provisioning status
[root@hpc-master ~]#
[root@hpc-master ~]# lsdef -t osimage
rocky8.10-x86_64-install-compute (osimage)
rocky8.10-x86_64-install-service (osimage)
rocky8.10-x86_64-netboot-compute (osimage)
rocky8.10-x86_64-stateful-mgmtnode (osimage)
rocky8.10-x86_64-statelite-compute (osimage)
[root@hpc-master ~]#
[root@hpc-master ~]#
[root@hpc-master ~]# tabdump nodelist
#node,groups,status,statustime,appstatus,appstatustime,primarysn,hidden,updatestatus,updatestatustime,zonename,comments,disable
"login","login,all","booted","03-02-2026 10:29:57",,,,,,,,,,
"compute1","compute,all","booted","03-02-2026 10:18:00",,,,,,,,,,
"hpc-master","__mgmtnode",,,,,,,,,,
"compute2","compute","booted","03-02-2026 10:27:21",,,,,,,,,,
"storage","storage","booted","03-02-2026 10:31:34",,,,,,,,,,
[root@hpc-master ~]#
```

5. Connectivity Tests

5.1 SSH Connectivity via xdsh

Command used:

```
xdsh compute1,compute2,login whoami
```

Output:

```
compute1: root
compute2: root
login: root
```

All three nodes respond to xdsh as root — SSH key trust is correctly established between hpc-master and all managed nodes.

5.2 Ping Tests

Commands used:

```
ping -c 2 compute1
ping -c 2 compute2
```

Results:

Node	IP	Packets Sent	Packets Received	Packet Loss	Avg RTT
compute1	192.168.100.13	2	2	0%	1.202 ms
compute2	192.168.100.14	2	2	0%	0.583 ms

Both compute nodes are reachable with zero packet loss.

```
[root@hpc-master ~]# #connectivity tests
[root@hpc-master ~]#
[root@hpc-master ~]# xdsh compute1,compute2,login whoami

compute1: root
compute2: root
login: root
[root@hpc-master ~]#
[root@hpc-master ~]#
[root@hpc-master ~]# ping -c 2 compute1
PING compute1.hpc.local (192.168.100.13) 56(84) bytes of data.
64 bytes from compute1.hpc.local (192.168.100.13): icmp_seq=1 ttl=64 time=0.537 ms
64 bytes from compute1.hpc.local (192.168.100.13): icmp_seq=2 ttl=64 time=1.87 ms

--- compute1.hpc.local ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.537/1.202/1.868/0.666 ms
[root@hpc-master ~]# ping -c 2 compute2
PING compute2.hpc.local (192.168.100.14) 56(84) bytes of data.
64 bytes from compute2.hpc.local (192.168.100.14): icmp_seq=1 ttl=64 time=0.540 ms
64 bytes from compute2.hpc.local (192.168.100.14): icmp_seq=2 ttl=64 time=0.626 ms

--- compute2.hpc.local ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.540/0.583/0.626/0.043 ms
[root@hpc-master ~]#
```

Overall Status: Cluster fully operational